BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Testing and Test Data

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does anticipating misuse mean in defensive design?

2. Merge sorted [1, 5] and [4, 6]. Which result is ascending?
   A) [1, 5, 4, 6]
   B) [4, 6, 1, 5]
   C) [6, 5, 4, 1]
   D) [1, 4, 5, 6]

3. "A comparison operator must always produce a text paragraph." Correct this. Explain why.

4. A school shares files and printers on one site. "A LAN must span several countries." Correct this. Why does a LAN fit here?

Today's objectives

Explain the core ideas of testing and test data using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Normal test
A typical value within permitted range.

Boundary test
A value at the smallest or largest allowed limit.

Invalid test
A value of the right type but outside allowed rules.

Erroneous test
Input of the wrong type that the program should reject safely.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Normal test
A typical value within permitted range.

Boundary test
A value at the smallest or largest allowed limit.

Invalid test
A value of the right type but outside allowed rules.

Erroneous test
Input of the wrong type that the program should reject safely.

I Do - Worked model

Create a test table for an integer score from 0 to 100 inclusive.

- Normal 50 should be accepted.
- Boundary 0 and 100 should both be accepted.
- Invalid -1 and 101 should both be rejected.
- Erroneous 'hello' should produce a safe error, not be accepted.

We Do - Guided practice

1. Plan tests for a quantity between 1 and 10.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Choose a negative test value to check that the program rejects it.

```
fileMB = 700
if fileMB >= 0 AND fileMB <= 1000 then
print("Accepted")
else
print("Rejected")
endif
```

You Do - Independent exam questions

Q2. Identify two boundary tests for valid ages 13-18. [2 marks]

Q3. Distinguish invalid and erroneous data for a numeric mark. [4 marks]

Q4. State why expected outcomes belong in a test plan. [2 marks]

Exit Ticket - True or False?

For range 1-10, 11 is a valid boundary value.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Planning for invalid, unexpected or malicious use and handling it safely.

2. D - [1, 4, 5, 6]

3. A Boolean result: true or false.

4. A LAN connects devices in a small area, such as one school site. A WAN links networks over a large area. Internet access alone does not make a LAN a WAN.

I Do - Explain the model

Normal 50 should be accepted.

Boundary 0 and 100 should both be accepted.

Invalid -1 and 101 should both be rejected.

Erroneous 'hello' should produce a safe error, not be accepted.

We Do - Model answers

1. An erroneous text input is 'ten'; expected outcome is rejection.

Code First - Answer

```
fileMB = -1
if fileMB >= 0 AND fileMB <= 1000 then
print("Accepted")
else
print("Rejected")
endif
```

Expected output: Rejected

You Do - Self-mark

Q2: 13 and 18.

Q3: Invalid: numeric 101 outside 0-100; erroneous: text 'apple' where an integer is expected.

Q4: They allow actual results to be compared with requirements and faults detected.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.